Smart Routine Generator 🚀
A specialized productivity tool for engineering students to manage their daily tasks efficiently.

📌 Project Overview
The **Smart Routine Generator** is a Python-
based desktop application developed as part of the **Programming Fundamentals (Level 100)** course at Mansoura National University.

The application provides a user-friendly Graphical User Interface (GUI) to help students organize their academic and personal tasks, set time slots, and ensure their schedule is saved for future reference.

✨ Key Features
User-Friendly GUI: Built using Python's `Tkinter` library.
Data Persistence: Uses **File Handling** to save and load your routine from a text file (`routine.txt`), so your data isn't lost when the app closes.
Real-time Interaction: Add tasks instantly and clear your history with a single click.
Input Validation:** Ensures no empty tasks are added to your schedule.

👥 Team Members
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🛠️ Technologies Used
Language: Python 3.
GUI Framework: Tkinter
Storage: File Handling (TXT)
Version Control: GitHub

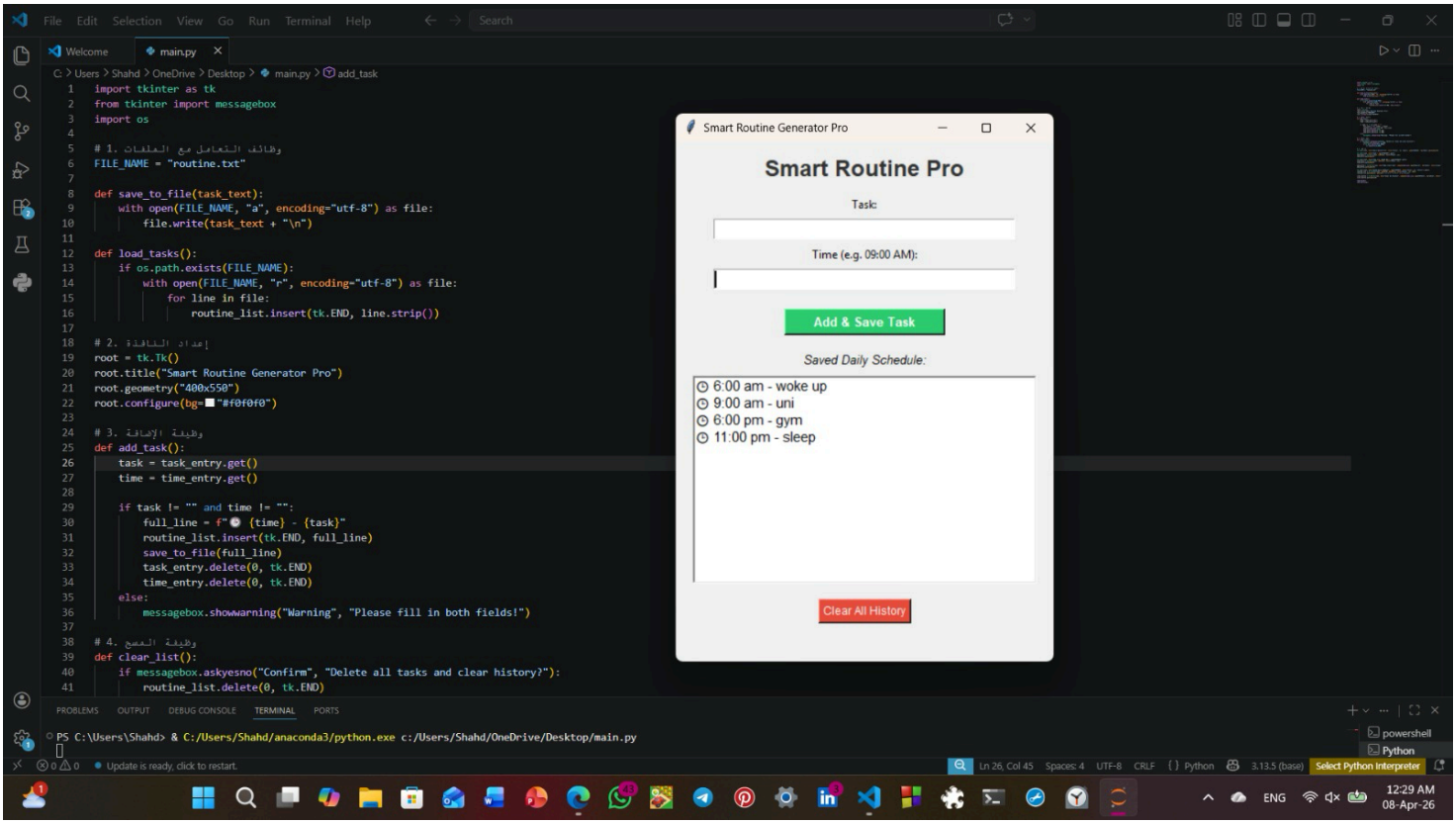
🚀 How to Run the App
1. **Clone the repository:**
   ```bash
   git clone [https://github.com/shahdmo109-sys/Smart-Routine-Generator]
   ```
2. **Navigate to the project directory:**

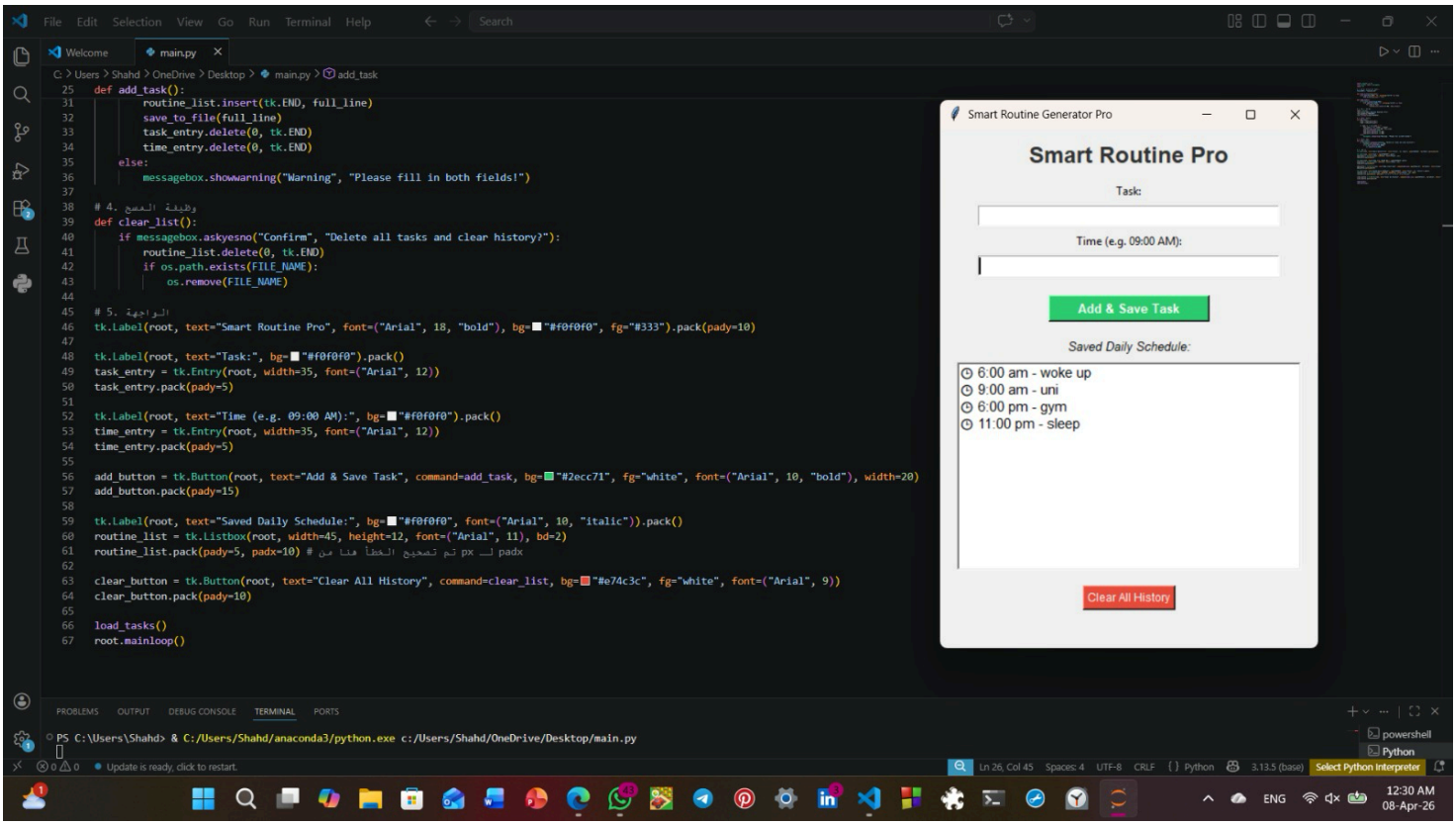
```bash
cd Smart-Routine-Generator
```

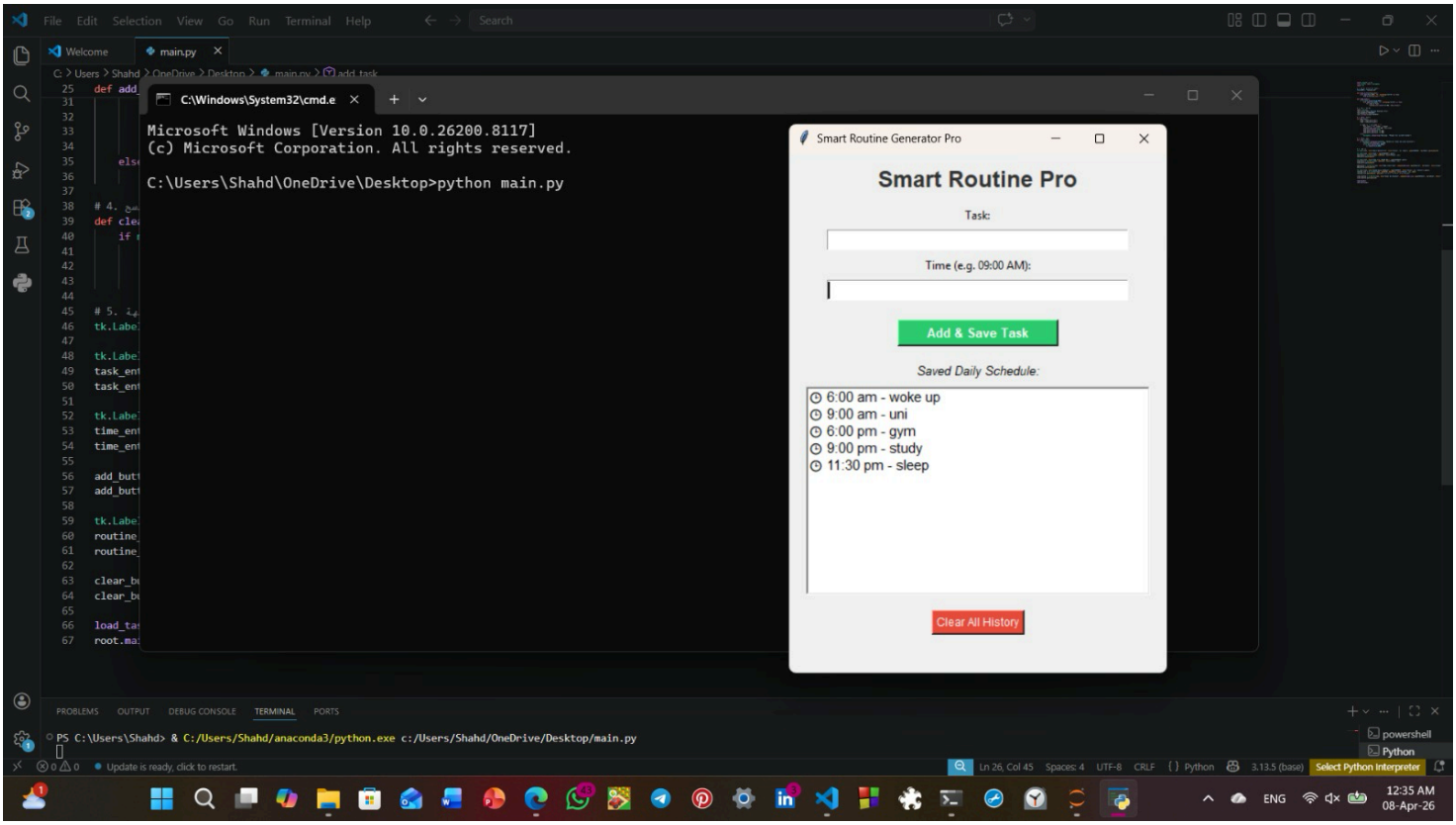
3. **Run the application:**

```bash
python main.py
```

📸 Screenshots




def add_task():
routine_list.insert(tk.END, full_line)
save_to_file(full_line)
task_entry.delete(0, tk.END)
time_entry.delete(0, tk.END)
else:
messagebox.showwarning("Warning", "Please fill in both fields!")
# 4. وظيفة المسح
def clear_list():
if messagebox.askyesno("Confirm", "Delete all tasks and clear history?"):
routine_list.delete(0, tk.END)
if os.path.exists(FILE_NAME):
os.remove(FILE_NAME)
# 5. الواجهة
tk.Label(root, text="Smart Routine Pro", font=("Arial", 18, "bold"), bg="#f0f0f0", fg="#333").pack(pady=10)
tk.Label(root, text="Task:", bg="#f0f0f0").pack()
task_entry = tk.Entry(root, width=35, font=("Arial", 12))
task_entry.pack(pady=5)
tk.Label(root, text="Time (e.g. 09:00 AM):", bg="#f0f0f0").pack()
time_entry = tk.Entry(root, width=35, font=("Arial", 12))
time_entry.pack(pady=5)
add_button = tk.Button(root, text="Add & Save Task", command=add_task, bg="#2ecc71", fg="white", font=("Arial", 10, "bold"), width=20)
add_button.pack(pady=15)
tk.Label(root, text="Saved Daily Schedule:", bg="#f0f0f0", font=("Arial", 10, "italic")).pack()
routine_list = tk.Listbox(root, width=45, height=12, font=("Arial", 11), bd=2)
routine_list.pack(pady=5, padx=10) # تم تصحيح الخطأ هنا من padx لـ px
clear_button = tk.Button(root, text="Clear All History", command=clear_list, bg="#e74c3c", fg="white", font=("Arial", 9))
clear_button.pack(pady=10)
load_tasks()
root.mainloop()
Smart Routine Generator Pro
Smart Routine Pro
Task:
Time (e.g. 09:00 AM):
Add & Save Task
Saved Daily Schedule:
6:00 am - woke up
9:00 am - uni
6:00 pm - gym
11:00 pm - sleep
Clear All History
PS C:\Users\Shahd> & C:/Users/Shahd/anaconda3/python.exe c:/Users/Shahd/OneDrive/Desktop/main.py


C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Shahd\OneDrive\Desktop>python main.py
Smart Routine Generator Pro
Smart Routine Pro
Task:
Time (e.g. 09:00 AM):
Add & Save Task
Saved Daily Schedule:
6:00 am - woke up
9:00 am - uni
6:00 pm - gym
9:00 pm - study
11:30 pm - sleep
Clear All History
PS C:\Users\Shahd> & C:/Users/Shahd/anaconda3/python.exe c:/Users/Shahd/OneDrive/Desktop/main.py

routine.txt

File Edit View

```
6:00 am - woke up
9:00 am - uni
6:00 pm - gym
9:00 pm - study
11:30 pm - sleep
```

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